

# A Noise-Tolerant Algorithm for Robot-Sensor Calibration Using a Planar Disk of Arbitrary 3-D Orientation

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**Abstract**—In a 3-D scanning task, a robot-sensor system controls a robotic arm to move a laser sensor. In order to align the coordinate system of the robotic arm and laser sensor, prior calibration is required to derive the transformation between both coordinate systems. This paper proposes a new calibration method in three steps: manual data collection, sensing data calculation, and transformation solution. First, at least four data are required to be collected by the user. The sensing data are then calculated from the collected data and adopted to provide the desired transformation. The proposed algorithm has two features: arbitrary placement of planar disk and noise tolerant. Using a planar disk, the algorithm will automatically derive the angular relationship between the disk and the sensor plane, enabling arbitrary orientation placement. Noise tolerant is guaranteed by fitting ellipses during the sensing data calculation and using a single set of sensing data in transformation solution. Experiments and comparisons are given to demonstrate the efficiency of the proposed calibration algorithm.

**Note to Practitioners**—This paper was motivated by the problem of calibrating a laser sensor and a positioning device (robot arm, CMM, etc.) in a robust and fast manner. Specifically, the calibration is to derive the transformation by aligning the sensors coordinate system to the positioning devices coordinate system. The proposed calibration procedure consists of two parts: manual data collection and automatic transformation calculation. During manual data collection, users only need to select four different data; whereby each data contains of two positions with the same orientation. Then, the desired transformation will be derived automatically. The calibration is designed in an efficient and robust way whereby: 1) data collection is done using a simple planar disk placed in arbitrary orientations; 2) minimum human interaction required; 3) tolerant to noise in the sensor data; and 4) easy implementation by following a proven and standard protocol.

**Index Terms**—Calibration disk, elliptic fitting, noise tolerant, robot-sensor calibration.

## I. INTRODUCTION

**C**ALIBRATION is to determine the relationship among different measuring devices [1]–[3]. In any tool/flange or hand/eye integrated system involving moving sensor [4]–[6],

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finding the geometry transformation between two coordinate systems is crucial for achieving accurate 3-D geometry. Such calibration is widely used in reverse engineering [7], robot-assisted medical applications [8], visually guided robot grinding [9], and robotic visual inspection [10]. Calibration methods vary in different aspects, such as calibration targets, data acquisition, and ways to derive the transformation.

All calibrations require some form of calibration targets, such as 2-D boards or 3-D sphere-shaped balls. Usually, different types of sensors use different calibration targets. For example, cameras uses 2-D calibration boards [5] while the profile laser sensors use 3-D calibration balls [11]–[15]. In comparison, a 2-D disk is cheaper and convenient, but has not been used as calibration target for laser sensors. The challenge is to account for the orientation of the 2-D disk in the base coordinate system during the calibration, and the estimation of the orientation has not been discussed before. The fixed-point calibration algorithm was introduced in [16] where the calibration target is a fixed cross-point printed on a 2-D disk rather than the entire disk.

A profile laser sensor projects a laser line on the object and provides a 2-D profile in the sensor plane. Different calibration methods have been developed for profile laser sensors, which require users to manually move the laser line to certain positions to acquire the sensing data. For example, the fixed-point calibration algorithm [16] required the users to manually move the laser line until it passes through the specific fixed point. On the other hand, the ball-based calibration algorithms [11]–[14] perform better as users can project a laser line on any part of the ball. Normally, one set of sensing data is required for the desired transformation [15]. But in two-step calibration methods [11]–[14], users need to acquire two different sets of sensing data: one set uses robotic arm with a fixed orientation while the other set uses robotic arm with different orientations. The two sets of data are used to solve the rotation and translation aspects of the transformation, respectively.

Once the sensing data for calibration are obtained, the desired transformation is solved in different manners. Traditional calibration methods are by means of homogeneous transformation. The well-known hand/eye calibration was formulated by Shiu and Ahmad [4] and Tsai and Lenz [5] by solving a homogeneous transform equation using nonlinear minimization [17], dual quaternions [18], etc. Another solution is the two-step calibration algorithm, which derives the rotation and translation portions separately [11]–[14]. Two

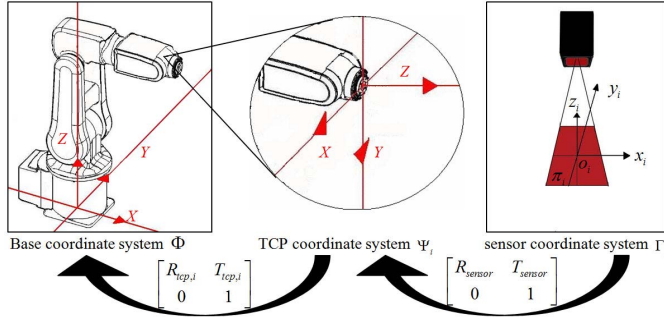


Fig. 1. Three different coordinate systems.

limitations arise from the two-step algorithms. First, noise in the sensing data and errors caused in the first step will be propagated to the second step [19]. Second, when more sensing data are required, the amount of intensive manual labor work is also increased. Minimal manual work in data collection for the calibration is desired for better accuracy and faster processing.

For the calibration of a profile laser sensor mounted onto a robotic arm, this paper targets to develop a new noise-tolerant calibration algorithm with a simple calibration target, easy data acquisition procedure, and lesser sensing data compared with existing techniques. A planar disk is chosen as the calibration target. To acquire the sensing data for the calibration, users can project the laser line onto the 2-D disk without the need to pass through any specific points, including the disk center. The proposed algorithm will be based on a single set of sensing data. Not only the amount of manual effort in data collection is reduced, but the amount of error propagation is also reduced. Most importantly, the calibration algorithm is robust to noise in laser sensing data and mechanical vibrations during arm movement.

This paper is structured as follows. Section I introduces the research background, and Section II gives the preliminaries. After that, the proposed calibration method is presented in Section III. Section IV proves the formulas used during calibration. Experimental results and comparisons are given in Section V. Finally, Section VI concludes this paper.

## II. SYSTEM AND NOTATIONS

### A. Robot-Sensor System

A robot-sensor system uses a robotic arm connected to a profile sensor to perform a 3-D scanning on target objects. The coordinate systems are shown in Fig. 1. The robotic arm has a fixed base coordinate system  $\Phi$ . At the  $i$ th position and orientation, the tool center point (TCP; tool0) coordinate system and the sensor coordinate system are denoted as  $\Psi_i$  and  $\Gamma_i$ , respectively. Notations are as follows.

- 1)  $R_{tcp,i}$  is the rotation matrix from  $\Psi_i$  to  $\Phi$ .
- 2)  $T_{tcp,i}$  is the translation vector from  $\Psi_i$  to  $\Phi$ .
- 3)  $R_{sensor}$  is the rotation matrix from  $\Gamma_i$  to  $\Psi_i$ .
- 4)  $T_{sensor}$  is the translation vector from  $\Gamma_i$  to  $\Psi_i$ .

Since the profile sensor is mounted on TCP, for TCP at different positions with different orientations,  $R_{sensor}$  and  $T_{sensor}$  remain unchanged. The calibration aims to derive  $R_{sensor}$  and

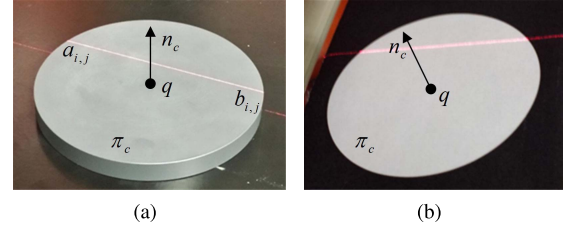


Fig. 2. Planar disks with 100 mm diameter. (a) Planar disk on the top of a cylinder. (b) Planar white disk on a black paper.

$T_{sensor}$  based on the data from the robotic arm and the sensor. At the  $i$ th position and orientation, the profile sensor generates a planar profile in the sensor plane  $\pi_i$ . Its coordinate system  $\Gamma_i = o_i x_i y_i z_i$  is defined, such that the origin  $o_i$  locates in the plane  $\pi_i$  and  $y_i$  is the normal of  $\pi_i$ . Following such notations, the second coordinate of the 2-D profile from the sensor is fixed as zero, i.e.,  $y = 0$  for the profile data.

### B. Calibration Target

A calibration target is an object where laser lights will be shone during data collection. Usually, 3-D balls are preferred, because the sensor has a uniform view from any position and orientation [15]. A ball with known radius has been used as the calibration target for robot-laser sensor calibration [11]–[14]. Different from previous methods, we plan to calibrate the system using a 2-D disk other than a 3-D object. Fig. 2 presents two forms of planar disks with known radius.

Profile data from the laser-scanning sensor can be divided into two types: chord data and nonchord data. On each profile, a chord is the line segment inside the disk, and the remnant data are nonchord data. Data preprocessing can remove the nonchord data in the profile. In Fig. 2(a), nonchord data and chord data are on two different lines. Thus, nonchord data can be easily removed based on the profile's geometry. In Fig. 2(b), nonchord data in black regions can be removed based on the profile's image. In this paper, experiments will be conducted using the 2-D disk shown in Fig. 2(a) and chords are derived using the split-and-merge algorithm [20]. Take the following notation.

- 1)  $\pi_c$  is the plane where the disk locates.
- 2)  $q$  is the center of the disk.
- 3)  $r_c$  is the radius of the disk.
- 4)  $n_c$  is the normal of  $\pi_c$ .
- 5)  $c_{i,j} = [a_{i,j}, b_{i,j}]$  is a chord with two tips  $a_{i,j}$  and  $b_{i,j}$ .

The new calibration method is dependent of  $q$  and  $n_c$ . Namely, the position and the orientation of the planar disk do not affect the calibration. Therefore, the planar disk can be placed at any position with any orientation for the calibration.

### C. Notations

For TCP at a position  $T_{tcp,i}$  with orientation  $R_{tcp,i}$ , the sensor plane is  $\pi_i$  with the corresponding sensor coordinate system  $\Gamma_i$ . For disk-based calibration,  $n$  sensing data are required as

$$\{T_{tcp,i}, R_{tcp,i}, q^i, D_i^a, D_i^b\}, \quad i = 1, 2, \dots, n$$

where

$R_{\text{tcp},i}$  and  $T_{\text{tcp},i}$  TCP orientation and position for  $\pi_i$  passing through the disk center  $q$ ;  
 $q^i = (q_x^i, 0, q_z^i)$  coordinates of  $q$  in  $\Gamma_i$ ;  
 $D_i$  TCP offset with coordinates  $D_i^b$  in  $\Phi$ , and coordinates  $D_i^i$  in  $\Gamma_i$ .

Users can manually control the robotic arm and visually check the laser until it passes through the disk center  $q$ . However, due to human error and noise from the scanning system, it is difficult to achieve high accuracy. Thus, in our system, users only need to select a TCP orientation  $R_{\text{tcp},i}$  and two TCP positions. The system will automatically calculate the desired  $T_{\text{tcp},i}$ .

For any physical point  $p$ ,  $p^{xyz}$  marks its coordinates in the coordinate system  $xyz$ . Following this notation,  $p^b = p^{\text{base}}$  denotes the coordinates of  $p$  in the base coordinate system  $\Phi$ . A TCP position  $T_{\text{tcp},i}$  is always in the base coordinate system, i.e.,  $T_{\text{tcp},i} = T_{\text{tcp},i}^b$ . For the  $i$ th sensing data, when the sensor plane passes through the disk center, the sensor coordinate system is  $\Gamma_i = o_i x_i y_i z_i$ . Then,  $p^{o_i x_i y_i z_i}$  becomes the coordinates of  $p$  in  $\Gamma_i$ . During the calibration,  $\Gamma_i$  will be rotated to the rotated coordinate system  $\Gamma_{i,r}$ .

The point pair  $p^{\text{base}}$  and  $p^{o_i x_i y_i z_i}$  refers to the same physical point. Each pair is named as a conjugate pair. Multiple conjugate pairs are required to enable the calibration [15]. Referring to Fig. 1, for the  $i$ th sensing data, a conjugate pair  $q^b$  and  $q^i$  satisfies

$$\begin{bmatrix} q^b \\ 1 \end{bmatrix} = \begin{bmatrix} R_{\text{tcp},i} & T_{\text{tcp},i} \\ 0 & 1 \end{bmatrix} \begin{bmatrix} R_{\text{sensor}} & T_{\text{sensor}} \\ 0 & 1 \end{bmatrix} \begin{bmatrix} q^i \\ 1 \end{bmatrix}$$

i.e.

$$q^b = R_{\text{tcp},i} R_{\text{sensor}} q^i + R_{\text{tcp},i} T_{\text{sensor}} + T_{\text{tcp},i}. \quad (1)$$

In this equation, the coordinate  $q^b$  is unknown before the calibration, as the disk is freely placed in the scanning system. The rotation matrix  $R_{\text{tcp},i}$  can be read from the robotic arm. The translation  $T_{\text{tcp},i}$  and the coordinate  $q^i$  will be calculated according to the chords. The aim of the calibration is to obtain the rotation matrix  $R_{\text{sensor}}$  and the translation  $T_{\text{sensor}}$ .

### III. NOISE-TOLERANT CALIBRATION ALGORITHM

Fig. 3 is the flowchart for the calibration process, where

|                                                                   |                                                                                                                               |
|-------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| $n$                                                               | number of sensing data to be collected,                                                                                       |
| $m$                                                               | number of parallel chords to be scanned for the $i$ th sensing data;                                                          |
| $\tilde{T}_{\text{tcp},i}$ and $\tilde{\tilde{T}}_{\text{tcp},i}$ | two TCP positions manually selected for the $i$ th sensing data;                                                              |
| $s_i$                                                             | value 1 or $-1$ ;                                                                                                             |
| $c_{i,j}$                                                         | values are $m$ parallel chords scanned when TCP moves from $\tilde{\tilde{T}}_{\text{tcp},i}$ to $\tilde{T}_{\text{tcp},i}$ ; |
| $\pi_i$                                                           | sensor plane for TCP at the $i$ th position $T_{\text{tcp},i}$ with orientation $R_{\text{tcp},i}$ ;                          |
| $\alpha_i$                                                        | angle between the sensor plane $\pi_i$ and the disk plane $\pi_c$ .                                                           |

According to the flowchart, the calibration algorithm contains three parts: manual data collection, sensing data

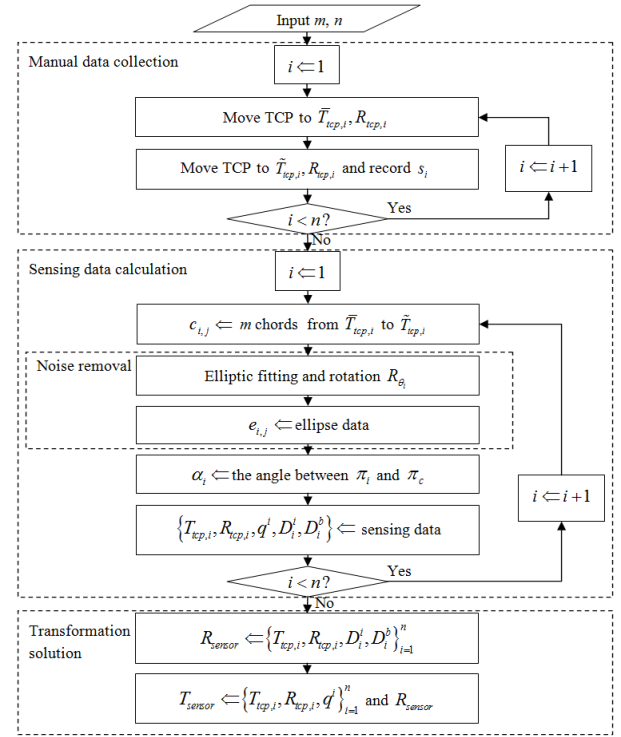


Fig. 3. Flowchart of the noise-tolerant calibration.

calculation, and transformation solution. For the  $i$ th sensing data, users only need to manually move TCP to two different positions  $\tilde{T}_{\text{tcp},i}$  and  $\tilde{\tilde{T}}_{\text{tcp},i}$  with the same orientation  $R_{\text{tcp},i}$ . During sensing data calculation,  $m$  chords are obtained and the  $i$ th sensing data  $\{T_{\text{tcp},i}, R_{\text{tcp},i}, q^i, D_i^i, D_i^b\}$  are calculated based on these chords. Finally, all sensing data are combined together to derive  $R_{\text{sensor}}$  and  $T_{\text{sensor}}$ . This section will detail the key formulas and steps in the flowchart. Section IV will show how to deduct these formulas.

#### A. Manual Data Collection

In each scan, the 3-D ball-based calibration methods [11]–[14] obtained a circular arc from the sensor and located the ball center directly from this arc. In our case, only one single chord can be obtained in each scan [Fig. 2(a)]. This chord alone is not sufficient to derive the coordinates of the disk center. Multiple chords are required in order to derive the  $i$ th sensing data (see details in Section IV-B). However, users do not need to manually scan each chord, which will be painful and time consuming. According to Fig. 3, users only need to manually select two different TCP positions  $\tilde{T}_{\text{tcp},i}$  and  $\tilde{\tilde{T}}_{\text{tcp},i}$  with a fixed orientation  $R_{\text{tcp},i}$ . With  $M_i = \tilde{\tilde{T}}_{\text{tcp},i} - \tilde{T}_{\text{tcp},i}$ , the scanning system can automatically translate TCP from  $\tilde{\tilde{T}}_{\text{tcp},i}$  by  $M_i/(m-1)$  for  $(m-1)$  times to obtain  $m$  chords  $\{c_{i,j}\}_{j=1}^m$  (the black chords in Fig. 4). To achieve a better calibration result,  $\tilde{\tilde{T}}_{\text{tcp},i}$  is selected, such that the TCP's moving direction  $M_i$  is neither parallel nor perpendicular to the calibration disk, that is

$$M_i \not\parallel \pi_c, M_i \not\perp \pi_c. \quad (2)$$

With the above-mentioned constraints, users can choose different  $\tilde{T}_{\text{tcp},i}$  and  $\tilde{\tilde{T}}_{\text{tcp},i}$ . The scanning can be within half of the disk [Fig. 4(a)] or the whole disk [Fig. 4(b)].



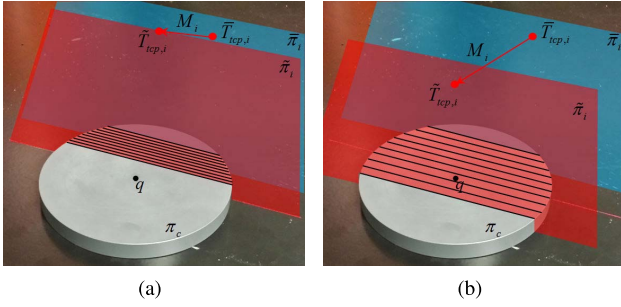


Fig. 4. Multiple chords scanning with  $m = 10$ . (a) Scan before  $q$ . (b) Scan over  $q$ .

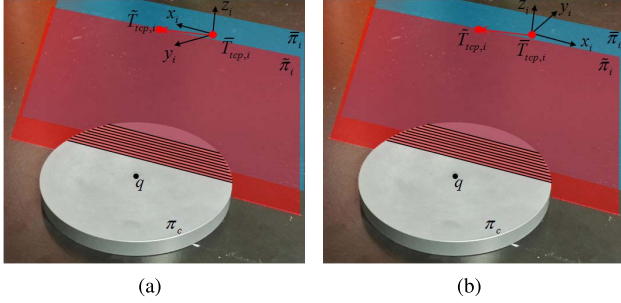


Fig. 5. Two possibilities. (a)  $s_i = 1$ . (b)  $s_i = -1$ .

Besides  $\tilde{T}_{tcp,i}$ ,  $\tilde{T}_{tcp,i}$ , and  $R_{tcp,i}$ , users also need to collect a value  $s_i$ , which is defined according to the TCP movement from  $\tilde{T}_{tcp,i}$  to  $\tilde{T}_{tcp,i}$ . If it follows the  $y^+$  direction in the sensor coordinate system [Fig. 5(a)], then  $s_i = 1$ . Otherwise, [Fig. 5(b)],  $s_i = -1$ .

Referring to Fig. 5, suppose  $\bar{\Gamma}_i$  is the sensor coordinate system in the sensor plane  $\bar{\pi}_i$ . If  $\tilde{T}_{tcp,i}$  is formulated in  $\bar{\Gamma}_i$  as  $\tilde{T}_{tcp,i}^{\bar{\Gamma}_i} = (T_{i,x}, T_{i,y}, T_{i,z})$ ,  $s_i$  can also be defined as

$$s_i = \begin{cases} 1, & T_{i,y} > 0, \\ -1, & T_{i,y} < 0. \end{cases} \quad (3)$$

The value  $s_i$  is necessary for ball-based calibration algorithms as well (see details in Section IV-A). The value  $s_i$  gives users more degrees of freedom during data collection. Users can easily have  $s_i = 1$  for all sensing data.

### B. Sensing Data Calculation

For the  $i$ th sensing data, sensor automatically scan each chord  $c_{i,j} = [a_{i,j}, b_{i,j}]$  [Fig. 2(a)] and provide 2-D chord data as

$$a_{i,j} = (a_{i,j,x}, 0, a_{i,j,z}), \quad b_{i,j} = (b_{i,j,x}, 0, b_{i,j,z}). \quad (4)$$

These chord data can be proved to be on the same ellipse in the plane  $\bar{\pi}_i$  (Fig. 6, see details in Section IV-C), and there may be noise within these chord data (see details in Section IV-D). Therefore, according to Fig. 3, these chord data are not directly used for sensing data calculation. Instead, ellipse fitting algorithm [21] is adopted for noise removal. The fitted ellipse will be used for calculating the desired sensing data.

There is an angle between  $x_i$  and  $\pi_c$  [Fig. 7(a)]. Applying a rotation matrix  $R_{\theta_i}$  in the sensor coordinate system  $\bar{\Gamma}_i$  in the sensor plane  $\bar{\pi}_i$ ,  $x_i y_i z_i$  becomes the rotated coordinate system  $\bar{x}_i \bar{y}_i \bar{z}_i$  [Fig. 7(b)]. The rotation matrix  $R_{\theta_i}$  can be selected such

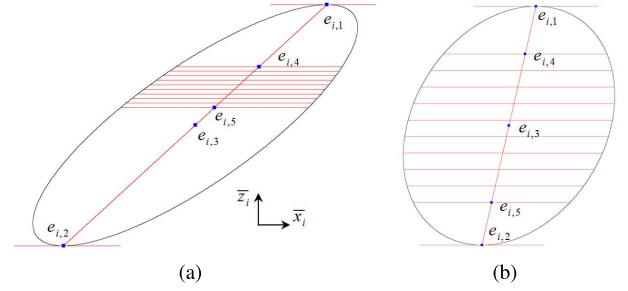


Fig. 6. Different ellipses in  $\bar{\pi}_i$ . (a) Fig. 4(a). (b) Fig. 4(b).

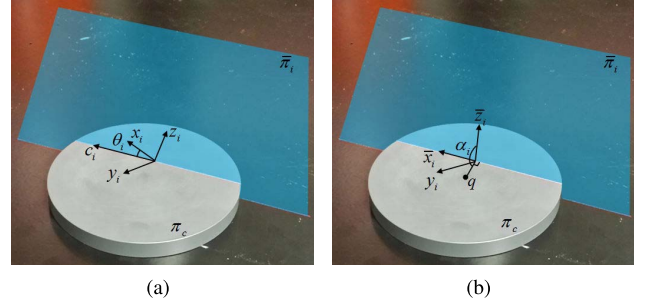


Fig. 7. Two coordinate systems in the sensor plane  $\bar{\pi}_i$ . (a)  $\bar{\Gamma}_i$ . (b)  $\bar{\Gamma}_{i,r}$ .

that  $\bar{x}_i \parallel \pi_c$ . In order to get the desired  $R_{\theta_i}$ , the chord direction for the  $i$ th sensing data is defined as [Fig. 7(a)]

$$c_i = \frac{1}{m} \sum_{j=1}^m (b_{i,j} - a_{i,j}).$$

Then, the rotation matrix  $R_{\theta_i}$  is defined by rotating  $x_i$  to  $c_i$  along  $y_i$  with angle  $\theta_i$ . After rotation, points  $\{e_{i,j}\}_{j=1}^5$  can be derived from the ellipse under the rotated coordinate system  $\bar{\Gamma}_{i,r}$  in the plane  $\bar{\pi}_i$  as follows (Fig. 6, see details in Section IV-E).

- 1)  $e_{i,1}$  and  $e_{i,2}$  are two ellipse points whose tangent directions are parallel to  $\bar{x}_i$  axis.
- 2)  $e_{i,3}$  is the center of the ellipse.
- 3)  $e_{i,4}/e_{i,5}$  is the intersection of the chord  $e_{i,1}e_{i,2}$  with the first/last chord.

The coordinates for  $e_{i,j}$  in  $\bar{\Gamma}_{i,r}$  and  $\bar{\Gamma}_i$  can be calculated as

$$e_{i,j}^{\bar{\Gamma}_{i,r}} = (e_{i,j,x}, 0, e_{i,j,z}) \quad (5)$$

$$e_{i,j}^{\bar{\Gamma}_i} = R_{\theta_i}^{-1}(e_{i,j,x}, 0, e_{i,j,z}). \quad (6)$$

Without loss of generality, as presented in Fig. 6,  $e_{i,1}$  and  $e_{i,2}$  can be selected, such that  $\overrightarrow{e_{i,1}e_{i,2}}$  and  $\overrightarrow{e_{i,4}e_{i,5}}$  are of the same direction.

For  $j = 1, 2, \dots, 5$ , denote

$$\eta_{i,j} = \frac{(e_{i,j} - e_{i,5}) \cdot (e_{i,4} - e_{i,5})}{|e_{i,4} - e_{i,5}|^2} \quad (7)$$

and

$$T_{i,j} = \eta_{i,j} \tilde{T}_{tcp,i} + (1 - \eta_{i,j}) \tilde{T}_{tcp,i}. \quad (8)$$

Then,  $T_{i,1}$  and  $T_{i,2}$  are two TCP positions where the sensor plane tangent to the disk, and  $T_{i,3}$  is the TCP position where the sensor plane passes through the disk center (see details in Section IV-F).

With (5) and (8), an angle  $\alpha_i$  can be obtained as [Fig. 7(b), see details in Section IV-G]

$$\alpha_i = \arccos \left( \frac{\|T_{i,1} - T_{i,2}\|^2 - \|e_{i,1} - e_{i,2}\|^2 - 4r_c^2}{4r_c(e_{i,1,z} - e_{i,2,z})} \right). \quad (9)$$

Then, the  $i$ th sensing data  $\{T_{\text{tcp},i}, R_{\text{tcp},i}, q^i, D_i^i, D_i^b\}$  are calculated as

$$T_{\text{tcp},i} = T_{i,3}, \quad (10)$$

$$q^i = R_{\theta_i}^{-1}(e_{i,3,x}, 0, e_{i,3,y}) \quad (11)$$

$$D_i^i = R_{\theta_i}^{-1}(e_{i,1,x} - e_{i,2,x}, 2r_c s_i \times \sin \alpha_i, e_{i,1,z} - e_{i,2,z} + 2r_c \cos \alpha_i) \quad (12)$$

$$D_i^b = T_{i,2} - T_{i,1}. \quad (13)$$

Details on how to derive these formulas will be presented in Section IV-G.

### C. Transformation Solution

Robot-sensor calibration is based on all sensing data  $\{T_{\text{tcp},i}, R_{\text{tcp},i}, q^i, D_i^i, D_i^b\}_{i=1}^n$ . Define two  $3 \times n$  matrices as

$$\begin{aligned} X &= [D_1^1, D_2^2, \dots, D_n^n] \\ Y &= [R_{\text{tcp},1}^{-1} D_1^b, R_{\text{tcp},2}^{-1} D_2^b, \dots, R_{\text{tcp},n}^{-1} D_n^b]. \end{aligned} \quad (14)$$

Apply singular value decomposition (SVD) [22] on the  $3 \times 3$  matrix  $H = XY^T$  as

$$H = USV^T \quad (15)$$

where  $U$  and  $V$  are  $3 \times 3$  orthogonal matrices while  $S$  is a  $3 \times 3$  diagonal matrix with nonnegative real numbers on the diagonal. Then, the rotation matrix can be derived as

$$R_{\text{sensor}} = VU^T. \quad (16)$$

Coupling with

$$\begin{aligned} R_i &= R_{\text{tcp},i} - R_{\text{tcp},i+1} \\ N_i &= R_{\text{tcp},i+1} R_{\text{sensor}} q^{i+1} - R_{\text{tcp},i} R_{\text{sensor}} q^i + T_{\text{tcp},i+1} - T_{\text{tcp},i} \end{aligned} \quad (17)$$

a  $3(n-1) \times 3$  matrix  $M$  and a  $3(n-1) \times 1$  matrix  $N$  can be defined as

$$M = \begin{bmatrix} R_1 \\ R_2 \\ \vdots \\ R_{n-1} \end{bmatrix}, \quad N = \begin{bmatrix} N_1 \\ N_2 \\ \vdots \\ N_{n-1} \end{bmatrix}. \quad (18)$$

Then, the translation vector can be formulated as

$$T_{\text{sensor}} = (M^T M)^{-1} (M^T N) \quad (19)$$

where  $M^T$  is the transpose of  $M$ . Section IV-H will detail how to formulate (16) and (19).

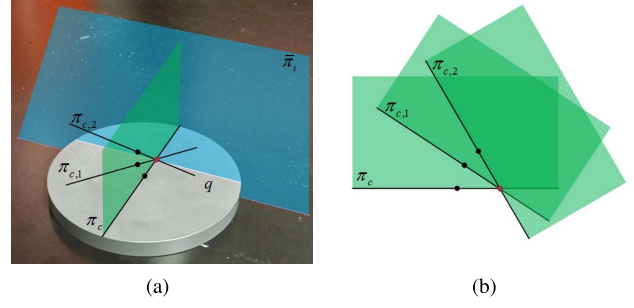


Fig. 8. Disk rotation along the chord. (a) Sensor scan one chord on the intersection between the disk plane and  $\pi_i$ . (b) Rotating the disk plane along the chord leads to the same chord from the sensor. (The green plane is perpendicular to both the disk plane and the chord.)

### D. Calibration Accuracy

In [13], the accuracy is estimated as the difference between the radius of the calibration target and the radius of the reconstructed target. In fact, as discussed during  $T_{\text{sensor}}$  calculation in [12], a ball can be reconstructed only if  $R_{\text{sensor}}$  is known. Thus,  $T_{\text{sensor}}$  will not affect the radius difference. In fact, radius difference only reflect the accuracy of  $R_{\text{sensor}}$ . Instead of the radius difference, the position difference should be used to estimate the calibration accuracy [23].

After calibration,  $\{T_{\text{tcp},i}, R_{\text{tcp},i}, q^i\}$  can be used to derive a disk center following (1) as:

$$q_i^b = R_{\text{tcp},i} R_{\text{sensor}} q^i + R_{\text{tcp},i} T_{\text{sensor}} + T_{\text{tcp},i}.$$

The disk center is estimated as

$$q^b = \sum_{i=1}^n q_i^b / n. \quad (20)$$

The calibration accuracy is

$$\max_k \|q^b - q_k^b\|.$$

The calibration accuracy is affected by the following factors:

- 1) the resolution and accuracy of the robotic arm;
- 2) the resolution and accuracy of the sensor;
- 3) the accuracy of the ellipse fitting on the training data.

## IV. ALGORITHM FORMULATION

During manual data collection,  $R_{\text{tcp},i}$ ,  $\bar{T}_{\text{tcp},i}$ ,  $\tilde{T}_{\text{tcp},i}$ , and  $s_i$  are collected. After that, chord data  $c_{i,j}$  are acquired from the sensor. This section will detail how the calibration algorithm is developed based on these data.

### A. Necessity of $s_i$

The sign  $s_i$  in (3) is recorded during manual data collection, and used in (12) during sensing data calculation. The ball-based calibration methods [11]–[14] also need such a value when formulating the ball center in the sensor coordinate system. Suppose the ball centers at  $c_b$  with radius  $r_b$ . In each scan, the sensor acquires a circular arc from the ball. Circle fitting algorithm is adopted to get the radius  $r_d$  and the center  $c_d = (c_{d,x}, 0, c_{d,z})$  of the circle in the sensor plane. Then,

the ball center can be formulated in the sensor coordinate system as

$$c_b = \left( c_{d,x}, s_i \sqrt{r_b^2 - r_d^2}, c_{d,z} \right)$$

where  $s_i$  can be  $-1$  or  $1$ . If the disk center locates on the  $y^+$  direction in the sensor coordinate system,  $s_i = 1$ . Otherwise,  $s_i = -1$ .

### B. Multiple Chords-Based Calculation

For TCP at any position, only a single chord is provided from the sensor [Fig. 8(a)]. If the chord passes through the disk center  $q$ , the coordinates  $q^i$  in the sensor coordinate system can be obtained as the chord center. Otherwise,  $q^i$  cannot be directly derived from the single chord. Consider a scenario where the robotic arm and the sensor are fixed while the circular disk rotates along the chord from  $\pi_c$  to  $\pi_{c,1}$  and  $\pi_{c,2}$  [Fig. 8(b)]. The chord data from the sensor remain unchanged, because the geometric distance from the chord to the sensor is fixed. Meanwhile, the coordinates  $q^i$  keep changing during the rotation. This means that a single chord alone is not sufficient to define  $q^i$ . The following sections will present that minimally two chords are required to locate the disk center. Moreover, to tolerate the noise during data acquisition, multiple chords should be collected.

### C. Chord Projection

Chords  $c_{i,j}$  in Fig. 4 are on the disk. When the sensor scans these chords, the sensor coordinate system moves. Thus, the readings from the sensor in (4) are in different coordinate systems. This section will prove that the chord data in (4) are on the same ellipse in the plane  $\bar{\pi}_i$ .

For the  $i$ th sensing data, referring to Fig. 4, TCP is translated by  $M_i/(m-1)$  for  $(m-1)$  times to move from  $\bar{T}_{tcp,i}$  to  $\bar{T}_{tcp,i}$ . Simultaneously,  $m$  parallel sensor planes  $\{\pi_{i,j}\}_{j=1}^m$  are derived satisfying  $\pi_{i,1} = \bar{\pi}_i$  and  $\pi_{i,m} = \bar{\pi}_i$ . In these sensor planes,  $m$  sensor coordinate systems  $\{\Gamma_{i,j} = o_{i,j}x_i y_i z_i\}_{j=1}^m$  also translate uniformly with (Fig. 9)

$$o_{i,j} = o_{i,1} + \frac{j-1}{m-1} M_i \quad (21)$$

where  $\bar{\Gamma}_i = \Gamma_{i,1} = o_{i,1}x_i y_i z_i$  is the sensor coordinate system in the sensor plane  $\bar{\pi}_i$ . Since there is only one translation between the two coordinate systems  $\Gamma_{i,1}$  and  $\Gamma_{i,j}$ , the coordinates of a vector in  $\Gamma_{i,1}$  equal to its coordinates in  $\Gamma_{i,j}$ , that is

$$M_i^{o_{i,1}x_i y_i z_i} = M_i^{o_{i,j}x_i y_i z_i}$$

and the coordinates of any point  $p$  in  $\Gamma_{i,1}$  and  $\Gamma_{i,j}$  satisfy

$$\begin{aligned} p^{o_{i,1}x_i y_i z_i} &= p^{o_{i,j}x_i y_i z_i} + \frac{j-1}{m-1} M_i^{o_{i,1}x_i y_i z_i} \\ &= p^{o_{i,j}x_i y_i z_i} + \frac{j-1}{m-1} M_i^{o_{i,j}x_i y_i z_i}. \end{aligned} \quad (22)$$

Reformulate sensor data in (4) for each chord  $c_{i,j}$  in  $\Gamma_{i,j}$  as [black parallel line segments in Fig. 9(a)]

$$c_{i,j}^{o_{i,j}x_i y_i z_i} = (c_{i,j,x}, 0, c_{i,j,z}). \quad (23)$$

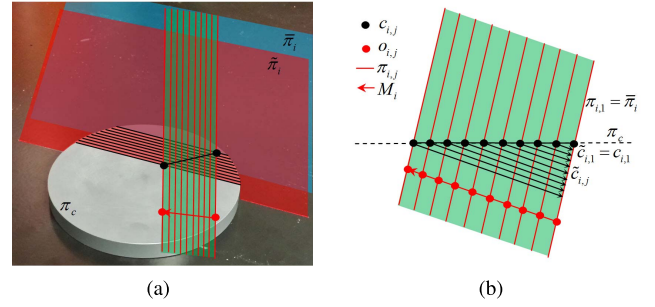


Fig. 9. Plane perpendicular to  $\pi_c$  and passing through  $o_{i,j}$ . (a) Plane in 3-D with red lines as its intersection with  $\pi_{i,j}$  and red arrow as the moving direction. (b) Plane in 2-D with black dots as  $c_{i,j}$ , red dots as  $o_{i,j}$ , black line as  $\pi_c$ , red lines as  $\pi_{i,j}$ , red arrow as  $M_i$ , and black arrows as projection directions.

Define a line segment  $\tilde{c}_{i,j}$  paralleling to  $c_{i,j}$  as

$$\tilde{c}_{i,j} = c_{i,j} - \frac{j-1}{m-1} M_i \quad (24)$$

and formulate it in  $\Gamma_{i,j}$  as

$$\tilde{c}_{i,j}^{o_{i,j}x_i y_i z_i} = c_{i,j}^{o_{i,j}x_i y_i z_i} - \frac{j-1}{m-1} M_i^{o_{i,j}x_i y_i z_i}. \quad (25)$$

On the other hand, substituting  $p = \tilde{c}_{i,j}$  into (22) gives

$$\tilde{c}_{i,j}^{o_{i,1}x_i y_i z_i} = \tilde{c}_{i,j}^{o_{i,j}x_i y_i z_i} + \frac{j-1}{m-1} M_i^{o_{i,j}x_i y_i z_i}. \quad (26)$$

Substituting (25) into (26) leads to

$$\tilde{c}_{i,j}^{o_{i,1}x_i y_i z_i} = c_{i,j}^{o_{i,j}x_i y_i z_i} = (c_{i,j,x}, 0, c_{i,j,z}). \quad (27)$$

Since the  $y$ -coordinate of  $\tilde{c}_{i,j}^{o_{i,1}x_i y_i z_i}$  is zero,  $\tilde{c}_{i,j}$  locates in the plane  $\bar{\pi}_i$ . In fact,  $\{\tilde{c}_{i,j}\}_{j=1}^m$  values are the projection of  $\{c_{i,j}\}_{j=1}^m$  to the plane  $\bar{\pi}_i$  following a direction parallel to  $M_i$  [black arrows in Fig. 9(b)]. Thus, the input sensor data  $\{(c_{i,j,x}, 0, c_{i,j,z})\}_{j=1}^m$  are not only the coordinates of chords  $\{c_{i,j}\}_{j=1}^m$  in different coordinate systems  $\{\Gamma_{i,j}\}_{j=1}^m$ , i.e.,  $\{c_{i,j}^{o_{i,j}x_i y_i z_i}\}_{j=1}^m$ , but also the coordinates of line segments  $\{\tilde{c}_{i,j}\}_{j=1}^m$  in the same coordinate system  $\bar{\Gamma}_i$ , i.e.,  $\{\tilde{c}_{i,j}^{o_{i,1}x_i y_i z_i}\}_{j=1}^m$ .

Geometrically, all  $\{c_{i,j}\}_{j=1}^m$  values are on the same disk. However, not all sensor data in (23) are on a disk, because they are under different coordinate systems  $\{\Gamma_{i,j}\}_{j=1}^m$ . With the constraints in (2), the projection of a circular disk to a plane following a direction parallel to  $M_i$  is an elliptic disk. Thus,  $\{(c_{i,j,x}, 0, c_{i,j,z})\}_{j=1}^m$  values are chords of this elliptic disk. The projected ellipse in  $\bar{\pi}_i$  can be acquired using ellipse fitting algorithm [21], and the chord data in (4) are on the projected ellipse.

### D. Noise Analysis in Sensor Data

In the sensor plane  $\bar{\pi}_i$ , the chords  $\{\tilde{c}_{i,j}\}_{j=1}^m$  in (27) center at  $\{s_{i,j}\}_{j=1}^m$ . According to the construction of  $e_{i,1}$  and  $e_{i,2}$ , all  $s_{i,j}$  values are geometrically on the chord  $e_{i,1}e_{i,2}$ . Formulate  $s_{i,j}$  in  $\bar{\Gamma}_i$  as  $s_{i,j}^{o_{i,1}x_i y_i z_i} = (s_{i,j,x}, 0, s_{i,j,z})$ . Then, ideally,  $(s_{i,j,x}, 0, s_{i,j,z})$  should be uniformly distributed on a line. However, in the actual scanned data,  $(s_{i,j,x}, 0, s_{i,j,z})$  may not be on the same line (Fig. 10). This is due to the noise

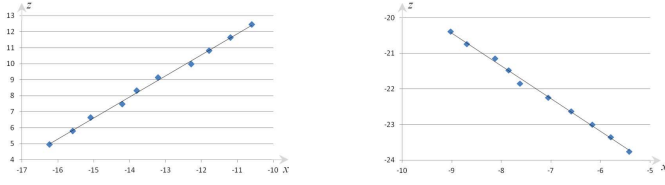


Fig. 10. Data  $(s_{i,j,x}, 0, s_{i,j,z})$  from the sensor do not lie exactly on the same line.

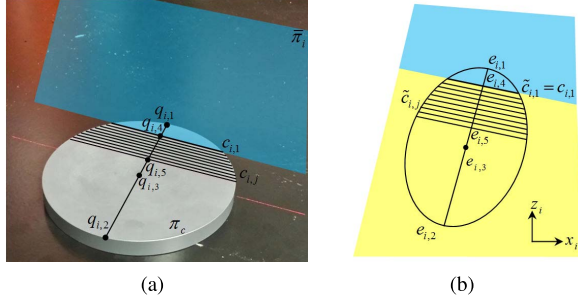


Fig. 11. Elliptic points via projection. (a)  $q_{i,j}$  in  $\pi_c$ . (b)  $e_{i,j}$  in  $\pi_i$ . (The blue area is above  $\pi_c$  and the yellow area is below  $\pi_c$ .)

introduced by the surface condition of the calibration target, the resolution of the sensor, the vibration of the robotic arm during sensing data acquisition, and so on.

To tolerate these noises, the proposed algorithm uses the fitted ellipse instead of the original chords for sensing data calculation. Theoretically, minimally two chords are required to enable the ellipse fitting. However, to better tolerate the noises,  $m \geq 100$  chords are collected for sensing data calculation (see details in Section V-B). For illustration purpose, figures in this paper adopts  $m = 10$ .

### E. Elliptic Points

During sensing data calculation, five elliptic points  $\{e_{i,j}\}_{j=1}^5$  on the projected ellipse instead of the raw sensor data are used. This section will introduce how these points are formulated as (5).

Referring to Fig. 11(a), on the circular disk, points

$$q_{i,j}, j = 1, 2, \dots, 5 \quad (28)$$

are defined in the plane  $\pi_c$ , such that the following holds.

- 1)  $q_{i,1}q_{i,2}$  is the diameter geometrically perpendicular to the chords  $\{c_{i,j}\}_{j=1}^m$ .
- 2)  $q_{i,3} = q$  is the disk center.
- 3)  $q_{i,4}$  and  $q_{i,5}$  are the intersections of  $q_{i,1}q_{i,2}$  to  $c_{i,1}$  and  $c_{i,m}$ , respectively.

Projecting  $q_{i,j}$  to the plane  $\pi_i$  following a direction parallel to  $M_i$  [Fig. 11(b)] leads to the projected points:

$$e_{i,j}, j = 1, 2, \dots, 5. \quad (29)$$

According to the definition, the following holds.

- 1) Ellipse chords  $\{\tilde{c}_{i,j}\}_{j=1}^m$  are the projections of circular chords  $\{c_{i,j}\}_{j=1}^m$ .
- 2) Ellipse chord  $e_{i,1}e_{i,2}$  is the projection of the diameter  $q_{i,1}q_{i,2}$ .
- 3) Center  $e_{i,3}$  is the projection of the center  $q = q_{i,3}$ .

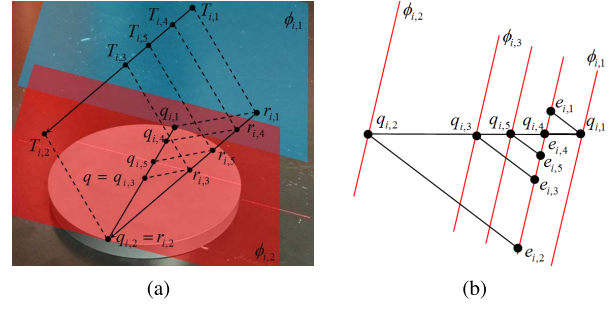


Fig. 12.  $T_{i,j}$  formulation. (a) Relationship among  $T_{i,j}$ ,  $r_{i,j}$ , and  $q_{i,j}$ . (b) Parallel projection from  $q_{i,j}$  to  $e_{i,j}$ .

- 4) Points  $e_{i,4}$  and  $e_{i,5}$  are the intersections of  $e_{i,1}e_{i,2}$  to  $\tilde{c}_{i,1}$  and  $\tilde{c}_{i,m}$ , respectively.

In the plane  $\pi_i$ , the chord  $e_{i,1}e_{i,2}$  passes through the centers of all ellipse chords parallel to  $\tilde{c}_{i,j}$ . Therefore, the tangent directions at  $e_{i,1}$  and  $e_{i,2}$  are parallel to  $\tilde{c}_{i,j}$  as well. After the rotation  $R_{\theta_i}$ , Fig. 11(b) becomes Fig. 6(a), and the tangent directions at  $e_{i,1}$  and  $e_{i,2}$  are parallel to the axis  $\bar{x}_i$ . In the rotated coordinate system  $\bar{\Gamma}_{i,r}$ , each point  $e_{i,j}$  is formulated in (5).

### F. TCP Positions

During data collection, users only need to manually collect two TCP positions  $\bar{T}_{tcp,i}$  and  $\tilde{T}_{tcp,i}$ . With the orientation  $R_{tcp,i}$  fixed, the proposed algorithm calculates other TCP positions where the sensor plane is passing through the disk center or tangent to the disk. Referring to Fig. 12(a), suppose that TCP positions  $\{T_{i,j}\}_{j=1}^5$  are linear combinations of  $\bar{T}_{tcp,i}$  and  $\tilde{T}_{tcp,i}$ . When TCP moves to  $T_{i,j}$ , the sensor plane is  $\phi_{i,j}$  and the sensor coordinate system is  $\Phi_{i,j}$ . With  $q_{i,j}$  defined in (28), the positions  $\{T_{i,j}\}_{j=1}^5$  are taken, such that the following holds.

- 1)  $\phi_{i,1}$  and  $\phi_{i,2}$  are tangent to the disk at the points  $q_{i,1}$  and  $q_{i,2}$ , respectively.
- 2)  $\phi_{i,3}$  passes through the disk center  $q = q_{i,3}$ .
- 3)  $\phi_{i,4} = \bar{\pi}_i$  passes through the point  $q_{i,4}$ .
- 4)  $\phi_{i,5} = \bar{\pi}_i$  passes through the point  $q_{i,5}$ .

This section is to prove that  $T_{i,j}$  satisfies (8).

Referring to Fig. 12(b), since  $\{e_{i,j}\}_{j=1}^5$  values are the parallel projection of  $\{q_{i,j}\}_{j=1}^5$ , similar with the chord projection in (27), the coordinates of  $q_{i,j}$  in the sensor coordinate system  $\Phi_{i,j}$  equal to the coordinates of  $e_{i,j}$  in the sensor coordinate system  $\Phi_{i,4}$ , that is

$$q_{i,j}^{\Phi_{i,j}} = e_{i,j}^{\bar{\Gamma}_i} = e_{i,j}^{\Phi_{i,4}}.$$

Since  $\phi_{i,4} = \bar{\pi}_i$  and  $\Phi_{i,4} = \bar{\Gamma}_i$ , coupling with (5) and (6), the coordinates of  $q_{i,j}$  in the coordinate system  $\Phi_{i,j}$  are

$$q_{i,j}^{\Phi_{i,j}} = e_{i,j}^{\Phi_{i,4}} = e_{i,j}^{\bar{\Gamma}_i} = R_{\theta_i}^{-1}(e_{i,1,x}, 0, e_{i,1,z}) \quad (30)$$

and its coordinates in the rotated coordinate system  $\Phi_{i,j,r}$  are

$$q_{i,j}^{\Phi_{i,j,r}} = e_{i,j}^{\Phi_{i,4,r}} = e_{i,j}^{\bar{\Gamma}_{i,r}} = (e_{i,1,x}, 0, e_{i,1,z}). \quad (31)$$

When TCP moves linearly, the sensor plane and the rotated coordinate system move linearly as well. Consider a point



$r_i$  in the sensor plane with fixed coordinates in the rotated coordinate system as  $(e_{i,1,x}, 0, e_{i,1,z})$ . Then, point  $r_i$  moves linearly as well. Referring to Fig. 12(a), when TCP moves to  $T_{i,j}$ ,  $r_i$  moves to  $r_{i,j}$  in  $\phi_{i,j}$  with

$$r_{i,j}^{\Phi_{i,j}} = R_{\theta_i}^{-1}(e_{i,1,x}, 0, e_{i,1,z}). \quad (32)$$

As such, all  $r_{i,j}$  values are on the same line, and

$$T_{i,j}r_{i,j} \parallel T_{i,1}r_{i,1}, \quad q_{i,j}r_{i,j} \parallel q_{i,1}r_{i,1}. \quad (33)$$

Coupling with (7),  $e_{i,j}$  can be formulated as

$$e_{i,j} = \eta_{i,j}e_{i,4} + (1 - \eta_{i,j})e_{i,5}.$$

Referring to Fig. 12(b), since  $\{e_{i,j}\}_{j=1}^5$  values are the parallel projection of  $\{q_{i,j}\}_{j=1}^5$ , line segments  $\{e_{i,j}e_{i,5}\}_{j=1}^4$  and  $\{q_{i,j}q_{i,5}\}_{j=1}^4$  are proportional segments, that is

$$\frac{\|e_{i,j}e_{i,5}\|}{\|e_{i,4}e_{i,5}\|} = \frac{\|q_{i,j}q_{i,5}\|}{\|q_{i,4}q_{i,5}\|}.$$

Thus

$$q_{i,j} = \eta_{i,j}q_{i,4} + (1 - \eta_{i,j})q_{i,5}.$$

Similarly, (33) leads to

$$\begin{aligned} r_{i,j} &= \eta_{i,j}r_{i,4} + (1 - \eta_{i,j})r_{i,5} \\ T_{i,j} &= \eta_{i,j}T_{i,4} + (1 - \eta_{i,j})T_{i,5} \end{aligned}$$

which proves (8).

### G. Sensing Data Formation

The  $i$ th sensing data are  $\{T_{\text{tcp},i}, R_{\text{tcp},i}, q^i, D_i^i, D_i^b\}$ , where  $R_{\text{tcp},i}$  is from the robotic arm. According to the definition of  $T_{i,j}$  in Section IV-F,  $T_{i,3}$  is the TCP position where the sensor plane passes through  $q$ . Thus,  $T_{\text{tcp},i}$  can be defined as (10). According to (30),  $q^i = q_{i,3}^{\Phi_{i,3}} = q_{i,3}^{\Phi_{i,j}} = e_{i,j}^{\Gamma_i}$ , which proves (11). The TCP offset  $D_i$  in the base coordinate system is given by (13). This section will show how to derive (12).

$\Gamma_i$  is the sensor coordinate system for the sensor plane passing through the disk center. According to the definition,  $\Phi_{i,3} = \Gamma_i$ . Suppose  $\Phi_{i,j,r}$  is the rotated coordinate system after applying  $R_{\theta_i}$  on  $\Phi_{i,j}$ . To derive the sensing data, different points need to be formulated in the same coordinate system  $\Phi_{i,1,r}$ .

Referring to (1), two different points  $r_{i,1}$  and  $q_{i,2}$  satisfy

$$\begin{aligned} r_{i,1}^b &= R_{\text{tcp},i}R_{\text{sensor}}r_{i,1}^{\Gamma_i} + R_{\text{tcp},i}T_{\text{sensor}} + T_{\text{tcp},i} \\ q_{i,2}^b &= R_{\text{tcp},i}R_{\text{sensor}}q_{i,2}^{\Gamma_i} + R_{\text{tcp},i}T_{\text{sensor}} + T_{\text{tcp},i}. \end{aligned} \quad (34)$$

Referring to Fig. 12(a), the TCP offset is

$$D_i = T_{i,2} - T_{i,1} = q_{i,2} - r_{i,1}. \quad (35)$$

Coupling with (34),  $D_i$  can be formulated as

$$D_i^b = q_{i,2}^b - r_{i,1}^b = R_{\text{tcp},i}R_{\text{sensor}}D_i^{\Gamma_i} = R_{\text{tcp},i}R_{\text{sensor}}D_i^i. \quad (36)$$

Since  $p^{\Gamma_i} = R_{\theta_i}^{-1}p^{\Gamma_{i,r}}$  holds for any point  $p$ , the above-mentioned equation becomes

$$D_i^b = R_{\text{tcp},i}R_{\text{sensor}}R_{\theta_i}^{-1}D_i^{\Gamma_{i,r}}. \quad (37)$$

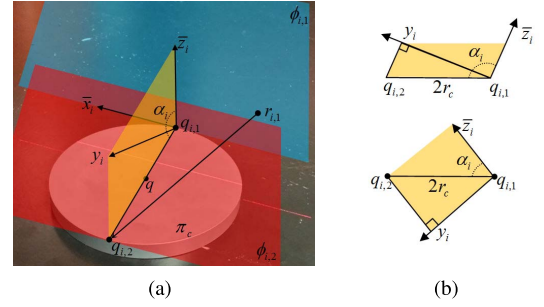


Fig. 13.  $\alpha_i$  formulation. (a) Rotated coordinate system in  $\Phi_{i,1}$ . (b) Different cases for  $\alpha_i$ .

The coordinates of a vector remain unchanged in  $\Phi_{i,1,r}$  and  $\Phi_{i,3,r}$ , because there is only a translation between the two coordinate systems. Substituting

$$D_i^{\Gamma_{i,r}} = D_i^{\Phi_{i,3,r}} = D_i^{\Phi_{i,1,r}} \quad (38)$$

into (37) gives

$$D_i^b = R_{\text{tcp},i}R_{\text{sensor}}R_{\theta_i}^{-1}D_i^{\Phi_{i,1,r}}. \quad (39)$$

In Fig. 13,  $\bar{x}_i\bar{y}_i\bar{z}_i$  is the rotated coordinate system, and  $\alpha_i$  is an angle from  $\bar{z}_i$  to  $\overrightarrow{q_{i,1}q_{i,2}}$ . Noted that  $\pi_c \parallel \bar{x}_i$ ,  $\bar{y}_i \perp \bar{x}_i$ ,  $\bar{z}_i \perp \bar{x}_i$  and  $q_{i,1}q_{i,2} \perp \bar{x}_i$ . The yellow plane spanned by  $\bar{y}_i, \bar{z}_i$  and  $q_{i,1}q_{i,2}$  is perpendicular to  $\pi_c$  and  $\bar{x}_i$  [Fig. 13(a)]. Thus,  $\alpha_i$  is an angle between  $\pi_c$  and  $\phi_{i,j}$ . Fig. 13(b) shows two possibilities for  $\alpha_i$ :  $\alpha_i \geq 90^\circ$  and  $\alpha_i < 90^\circ$ . According to (2), although TCP movement is not perpendicular to  $\pi_c$ ,  $\phi_{i,j}$  may be perpendicular to  $\pi_c$ , i.e.,  $\alpha_i = 90^\circ$ .

To derive  $\alpha_i$ , points  $q_{i,1}$ ,  $q_{i,2}$  and  $r_{i,1}$  also need to be formulated in the same coordinate system  $\Phi_{i,1,r}$ . Equations (31) and (32) give

$$q_{i,1}^{\Phi_{i,1,r}} = (e_{i,1,x}, 0, e_{i,1,z}) \quad (40)$$

$$r_{i,1}^{\Phi_{i,1,r}} = (e_{i,2,x}, 0, e_{i,2,z}). \quad (41)$$

According to the definition of  $q_{i,j}$  in (28) and  $s_i$  in (3), if  $s_i = 1$  [Fig. 13(b)],  $q_{i,2}$  is on the  $y^+$  direction of  $\Phi_{i,1,r}$ . Otherwise,  $q_{i,2}$  is on the  $y^-$  direction of  $\Phi_{i,1,r}$ . Thus, in the yellow plane perpendicular to  $\bar{x}_i$  (Fig. 13), the offset  $q_{i,2} - q_{i,1}$  depends on  $s_i$  as

$$(q_{i,2} - q_{i,1})^{\Phi_{i,1,r}} = 2r_c(0, s_i \cdot \sin \alpha_i, \cos \alpha_i).$$

Coupling with (40),  $q_{i,2}$  can be formulated as

$$\begin{aligned} q_{i,2}^{\Phi_{i,1,r}} &= q_{i,1}^{\Phi_{i,1,r}} + 2r_c(0, s_i \cdot \sin \alpha_i, \cos \alpha_i) \\ &= (e_{i,1,x}, 2r_c \cdot s_i \cdot \sin \alpha_i, e_{i,1,z} + 2r_c \cdot \cos \alpha_i). \end{aligned} \quad (42)$$

Combining (41) and (42),  $D_i$  defined in (35) can be formulated in  $\Phi_{i,1,r}$  as

$$\begin{aligned} D_i^{\Phi_{i,1,r}} &= q_{i,2}^{\Phi_{i,1,r}} - r_{i,1}^{\Phi_{i,1,r}} \\ &= (e_{i,1,x} - e_{i,2,x}, 2r_c \cdot s_i \\ &\quad \cdot \sin \alpha_i, e_{i,1,z} - e_{i,2,z} + 2r_c \cdot \cos \alpha_i). \end{aligned}$$

According to (38), (12) can be obtained from

$$D_i^i = D_i^{\Gamma_i} = R_{\theta_i}^{-1}D_i^{\Gamma_{i,r}} = R_{\theta_i}^{-1}D_i^{\Phi_{i,1,r}}.$$



TABLE I  
SENSING DATA FOR DISK-BASED CALIBRATION

|                     | $i = 1$                       | $i = 2$                       | $i = 3$                       | $i = 4$                       |
|---------------------|-------------------------------|-------------------------------|-------------------------------|-------------------------------|
| $T_{tcp,i}$         | (-222.966, -295.994, 360.786) | (-232.113, -298.165, 344.196) | (-374.932, -347.138, 338.836) | (-377.572, -373.611, 352.241) |
| $R_{tcp,i}^\dagger$ | (3.057, -0.103, -2.008)       | (3.115, -0.2, -2.269)         | (-2.981, 0.256, -2.491)       | (3.092, 0.208, 3.11)          |
| $q_i^\ddagger$      | (5.947, 0.5, 97)              | (-12.315, 0.24, 716)          | (9.328, 0.34, 277)            | (9.675, 0.15, 973)            |
| $D_i^{i\ddagger}$   | (0.042, 0.336, -0.941)        | (-0.146, 0.424, -0.894)       | (0.222, 0.589, -0.777)        | (-0.024, 0.674, -0.738)       |
| $D_i^{b\ddagger}$   | (0.181, -0.246, -0.952)       | (0.287, -0.294, -0.912)       | (0.391, -0.593, -0.704)       | (0.164, -0.631, -0.758)       |

<sup>†</sup> ZYX Euler angle.

<sup>‡</sup>  $D_i^{i\ddagger}$  and  $D_i^{b\ddagger}$  are normalized since  $\|D_i^{i\ddagger}\| = \|D_i^{b\ddagger}\|$ .

On the other hand, (39) gives  $\|D_i^b\| = \|D_i^{\Phi_{i,1,r}}\|$ , that is

$$\|T_{i,2} - T_{i,1}\|^2 = (e_{i,1,x} - e_{i,2,x})^2 + (e_{i,1,z} - e_{i,2,z})^2 + 4r_c^2 + 4r_c(e_{i,1,z} - e_{i,2,z})\cos\alpha_i.$$

According to (2), the sensor plane does not move parallel to  $\pi_c$ , i.e.,  $e_{i,1,z} \neq e_{i,2,z}$ . Therefore

$$\begin{aligned} \cos\alpha_i &= \frac{\|T_{i,2} - T_{i,1}\|^2 - (e_{i,1,x} - e_{i,2,x})^2 - (e_{i,1,z} - e_{i,2,z})^2 - 4r_c^2}{4r_c(e_{i,1,z} - e_{i,2,z})} \end{aligned} \quad (43)$$

which gives (9).

#### H. Algorithm Design

Robot-sensor calibration is to derive the desired  $R_{\text{sensor}}$  and  $T_{\text{sensor}}$  from all sensing data. Equation (36) gives

$$R_{\text{sensor}}D_i^i = R_{\text{tcp},i}^{-1}D_i^b, i = 1, 2, \dots, n$$

which can be reformulated into

$$\begin{aligned} R_{\text{sensor}}[D_1^1, D_2^2, \dots, D_n^n] \\ = [R_{\text{tcp},1}^{-1}D_1^b, R_{\text{tcp},2}^{-1}D_2^b, \dots, R_{\text{tcp},n}^{-1}D_n^b]. \end{aligned} \quad (44)$$

Coupling with notations in (14), the above-mentioned equations become

$$R_{\text{sensor}}X = Y. \quad (45)$$

The unknown  $R_{\text{sensor}}$  in (45) can be solved using the SVD [22], which gives the solution as (16).

On the other hand, according to (1), for  $i = 1, \dots, n-1$

$$\begin{aligned} q^b &= R_{\text{tcp},i+1}R_{\text{sensor}}q^{i+1} + R_{\text{tcp},i+1}T_{\text{sensor}} + T_{\text{tcp},i+1} \\ q^b &= R_{\text{tcp},i}R_{\text{sensor}}q^i + R_{\text{tcp},i}T_{\text{sensor}} + T_{\text{tcp},i}. \end{aligned}$$

The subtraction of the two equations provides

$$\begin{aligned} (R_{\text{tcp},i+1}R_{\text{sensor}}q^{i+1} + R_{\text{tcp},i+1}T_{\text{sensor}} + T_{\text{tcp},i+1}) \\ - (R_{\text{tcp},i}R_{\text{sensor}}q^i + R_{\text{tcp},i}T_{\text{sensor}} + T_{\text{tcp},i}) = 0 \end{aligned}$$

which gives

$$\begin{aligned} (R_{\text{tcp},i} - R_{\text{tcp},i+1})T_{\text{sensor}} &= R_{\text{tcp},i+1}R_{\text{sensor}}q^{i+1} \\ &\quad - R_{\text{tcp},i}R_{\text{sensor}}q^i + T_{\text{tcp},i+1} - T_{\text{tcp},i}. \end{aligned}$$

With notations in (17), the above-mentioned equations can be reformulated into

$$R_i T_{\text{sensor}} = N_i, \quad i = 1, 2, \dots, n-1$$

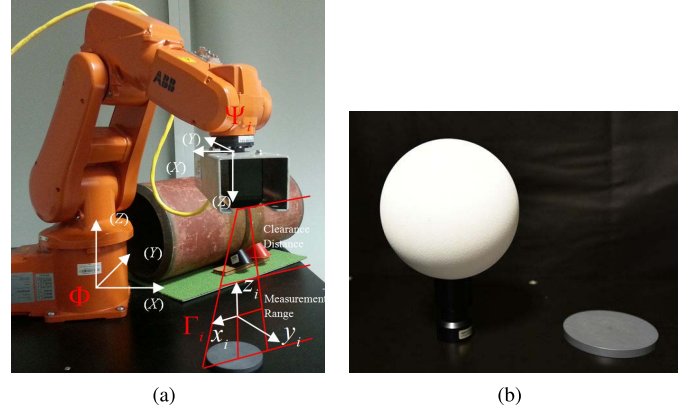


Fig. 14. Experimental setup. (a) Scanning system. (b) Calibration ball with radius 69.5 mm and calibration disk with radius 50 mm.

or equivalently

$$\begin{bmatrix} R_1 \\ R_2 \\ \vdots \\ R_{n-1} \end{bmatrix} T_{\text{sensor}} = \begin{bmatrix} N_1 \\ N_2 \\ \vdots \\ N_{n-1} \end{bmatrix}. \quad (46)$$

Substituting (18) into (46) leads to

$$MT_{\text{sensor}} = N. \quad (47)$$

The unknown  $T_{\text{sensor}}$  can be solved using the least-square method with the result presented in (19).

## V. EXPERIMENTAL RESULTS

### A. System Setup and Calibration

In the scanning system shown in Fig. 14(a), an LMI GOCATOR 2340 [24] laser profile sensor is mounted onto an ABB IRB 120 robotic arm [25]. The unit for the coordinate system is millimeter (mm).

Denote the manually collected TCP positions and orientations used for the calibration as

$$(\bar{R}_i, \bar{T}_i, \tilde{R}_i, \tilde{T}_i), i \in \{1, 2, 3, 4\}. \quad (48)$$

Table I lists the four sensing data derived using the scanning system in Fig. 14(a). The proposed noise-tolerant calibration algorithm derives the final transformation matrix as

$$\Lambda = \begin{bmatrix} 0.99996 & -0.00547 & -0.00716 & 6.0252 \\ -0.00544 & -0.99998 & 0.00288 & 54.18211 \\ -0.00718 & -0.00284 & -0.99997 & 364.25534 \\ 0 & 0 & 0 & 1 \end{bmatrix}. \quad (49)$$

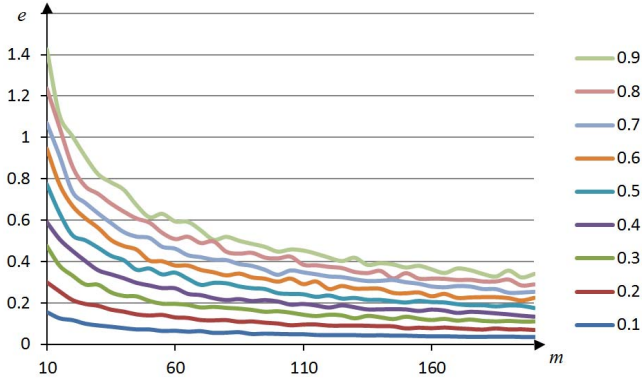


Fig. 15. Disk-based calibration accuracy with respect to  $\sigma$  and  $m$  [the horizontal coordinate is the number of chords  $m$ ; the vertical coordinate is the calibration accuracy  $e$ (mm), and curves in different colors correspond to different noise level  $\sigma$ (mm)].

Following Section III-D, the disk center in (20) is calculated as:

$$q^b = (-319.622, -304.086, 11.48) \quad (50)$$

and the calibration accuracy is estimated as

$$\max_{i=1}^4 \|q^b - q_i^b\| = 0.212 \text{ mm} \quad (51)$$

which is jointly affected by the  $x$ -resolution of the sensor, which is 0.1 mm, the robotic arm's positioning accuracy, and the ellipse fitting accuracy.

### B. Simulation and Validation

A simulated robotic-laser-scanning system is set up with a simulated calibration matrix  $\tilde{\Lambda} = \Lambda$  in (49). A virtual disk centered at  $\tilde{q}^b = q^b$  in (50) with orientation  $\tilde{n}_c^b = (0, 0, 1)$  is used for collecting simulation data. TCP data in (48) are adopted for the simulation. The simulated chord data  $\tilde{a}_{i,j}$  and  $\tilde{b}_{i,j}$  are collected as the intersections of the virtual disk and the moving laser planes. Then, the calibration based on the simulated chord data derives the transformation matrix  $\tilde{\Lambda}$  and the disk center  $\tilde{q}^b$ . Simulation results show that

$$\tilde{\Lambda} = \Lambda, \quad \tilde{q}^b = q^b$$

which validates the proposed calibration algorithm.

The above-mentioned simulation and validation do not consider the noise in the chord data. In fact, the real chord data  $a_{i,j}$  and  $b_{i,j}$  in (4) may contain noise introduced by the accuracy of the robotic arm, the  $x$ -resolution of the sensor, the accuracy of the chord segmentation, and so on. Assume that such noise is the Gaussian noise  $G(\sigma)$  with mean value 0 and standard deviation  $\sigma$ . A set of noise chord data from the simulated chord data is derived as

$$\begin{aligned} \tilde{a}_{i,j} &= \bar{a}_{i,j} + \frac{\tau_{i,j,1}}{\|\bar{b}_{i,j} - \bar{a}_{i,j}\|} (\bar{b}_{i,j} - \bar{a}_{i,j}) \\ \tilde{b}_{i,j} &= \bar{b}_{i,j} + \frac{\tau_{i,j,2}}{\|\bar{b}_{i,j} - \bar{a}_{i,j}\|} (\bar{b}_{i,j} - \bar{a}_{i,j}) \end{aligned} \quad (52)$$

where  $i = 1, 2, 3, 4$ ,  $j = 1, 2, \dots, m$ , and  $\tau_{i,j,k}$  values are Gaussian noise  $G(\sigma)$ .

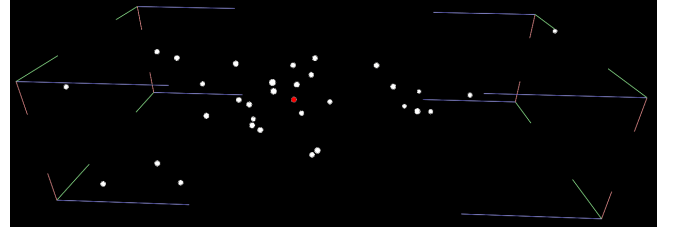


Fig. 16. Reconstructed disk centers in repeatability test (the white dots are disk centers from different tests; the red dot is the average of the 32 white dots; the size of the bounding box is 0.070 mm  $\times$  0.131 mm  $\times$  0.287 mm).

For each standard deviation  $\sigma \in \{0.1, 0.2, \dots, 0.9\}$ , 500 sets of noise chord data  $\{\tilde{a}_{i,j,l}, \tilde{b}_{i,j,l}\}_{l=1}^{500}$  are simulated according to (52). The calibration based on the  $l$ th set  $\{\tilde{a}_{i,j,l}, \tilde{b}_{i,j,l}\}$  gives a transformation matrix, a disk center  $\tilde{q}_l^b$ , and a calibration accuracy  $\varepsilon_l = \|\tilde{q}_l^b - \tilde{q}^b\|$  (the error in reconstructing the disk center). The standard deviation of  $\{\varepsilon_l\}_{l=1}^{500}$  is denoted as  $e$ , which indicates the calibration accuracy with respect to  $\sigma$  and  $m$ .

Fig. 15 shows the trend of the calibration accuracy  $e$  with respect to  $\sigma$  and  $m$ . In general, with  $m$  fixed,  $e$  increases as  $\sigma$  increases. For example, with  $m = 100$ , as  $\sigma$  increases from 0.1 to 0.9,  $e$  increases from 0.052 to 0.449. On the other hand, with  $\sigma$  fixed, bigger  $m$  gives better accuracy  $e$ . In this simulation, when  $m$  increases from 10 to 200 while  $\sigma$  fixed at 0.4, the accuracy  $e$  increases from 0.591 to 0.135 (the forth curve from the bottom in Fig. 15). According to curves in Fig. 15, bigger  $m$  provides better accuracy. For the experimental setup in Fig. 14(a), we adopt  $m = 100$ .

### C. Repeatability

To analyze the repeatability of the proposed calibration algorithm, with position and orientation of the disk fixed, calibration data are collected based on the same set of TCP data in (48). The data collection is repeated for 32 times providing 32 calibration results. The disk centers  $\{q_i^b\}_{i=1}^{32}$  are used to estimate the repeatability (the white dots in Fig. 16). For  $\{q_i^b\}_{i=1}^{32}$ , the mean is denoted as (the red dot in Fig. 16)

$$\dot{q} = \sum_{i=1}^{32} q_i^b / 32$$

and the standard deviation is

$$\sum_{i=1}^{32} \|q_i^b - \dot{q}\| / 32 = 0.071.$$

Thus, the repeatability of the proposed algorithm based on the same set of TCP data is 0.071 mm.

### D. Comparisons

1) *Different Methods*: The proposed method is different from the methods presented in [11]–[15] with different calibration targets and different calibration procedures (Table II). Balls had been used for the calibration between a CMM machine and a laser-stripe sensor [15]. The target is also to

TABLE II  
COMPARISON OF DIFFERENT CALIBRATION METHODS

|           | Calibration target | Data from a conjugate pair                    | Number of conjugate pairs | Method                                                                                                              | Required data for each conjugate pair                     |
|-----------|--------------------|-----------------------------------------------|---------------------------|---------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------|
| [15]      | Tetra-ball         | $\{q^b, q^i\}$                                | More than 3               | Constrained least-squares optimization                                                                              | $q^b$ and $q^i$                                           |
| [11]–[14] | Ball               | $\{T_{tcp,i}, R_{tcp,i}, q^i\}$               | At least 7                | At least 4 data for $R_{sensor}$ , and at least 3 data for $T_{sensor}$                                             | for $R_{sensor}$ : one arc<br>for $T_{sensor}$ : $m$ arcs |
| Ours      | Planar disk        | $\{T_{tcp,i}, R_{tcp,i}, q^i, D_i^i, D_i^b\}$ | At least 4                | $\{T_{tcp,i}, R_{tcp,i}, D_i^i, D_i^b\}$ for $R_{sensor}$ , and<br>$\{T_{tcp,i}, R_{tcp,i}, q^i\}$ for $T_{sensor}$ | $m$ chords                                                |

TABLE III  
SENSING DATA FOR BALL-BASED CALIBRATION

|         | $T_{tcp,i}$                   | $R_{tcp,i}^\dagger$      | $q^i$                      |
|---------|-------------------------------|--------------------------|----------------------------|
| $i = 1$ | (-215.407, -287.69, 559.998)  | (3.08, -0.25, -2.611)    | (-16.168, 43.629, -58.079) |
| $i = 2$ | (-179.4, -289.3, 550.402)     | (3.08, -0.25, -2.611)    | (15.42, 23.498, -57.636)   |
| $i = 3$ | (-187.799, -283.8, 502.902)   | (3.08, -0.25, -2.611)    | (23.695, 30.256, -10.06)   |
| $i = 4$ | (-133.8, -268.564, 099)       | (3.08, -0.25, -2.611)    | (60.036, 19.653, -84.071)  |
| $i = 5$ | (-317.98, -296.293, 567.379)  | (-3.074, -0.051, -2.689) | (-26.706, 34.022, -52.304) |
| $i = 6$ | (-290.941, -319.017, 496.149) | (3.116, -0.013, -3.027)  | (12.521, 45.942, 16.945)   |
| $i = 7$ | (-275.77, -326.73, 480.747)   | (3.09, -0.136, 3.037)    | (-8.488, 54.055, 29.974)   |
| $i = 8$ | (-217.283, -327.968, 569.943) | (2.994, -0.249, -2.467)  | (-50.322, 37.241, -64.803) |

$^\dagger$  ZYX Euler angle.

find out the transformation matrix

$$H = \begin{bmatrix} R_{sensor} & T_{sensor} \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} t_{11} & t_{12} & t_{13} & p_1 \\ t_{21} & t_{22} & t_{23} & p_2 \\ t_{31} & t_{32} & t_{33} & p_3 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

Multiple conjugate pairs  $\{q^b, q^i\}$  are collected and used to formulate a constrained least-squares optimization problem with augmented objective function

$$F = \sum_{i=1}^n |q^b - H \cdot q^i|^2 + \sum_{j=1}^3 \lambda_j [t_{j1}^2 + t_{j2}^2 + t_{j3}^2 - 1] + \lambda_4 [t_{12}t_{13} + t_{22}t_{23} + t_{32}t_{33}].$$

Solving such problem directly gives  $H$ . Theoretically, three conjugate pairs are sufficient. However, the constrained least-squares optimization problem may give more than one solution. Thus, more conjugate pairs are desired [15].

Another type of ball-based calibration methods [11]–[14] is different from that presented in [15]. First, the ball center  $q^b$  is fixed but is unknown, and data  $\{T_{tcp,i}, R_{tcp,i}, q^i\}$  are collected satisfying

$$q^b = R_{tcp,i} R_{sensor} q^i + R_{tcp,i} T_{sensor} + T_{tcp,i}.$$

This is different from [15], where  $\{q^b, q^i\}$  should be collected.  $R_{sensor}$  and  $T_{sensor}$  are solved separately. First, with TCP in translation mode, at least four data are collected and used to solve a set of linear equations for  $R_{sensor}$ . Then, as TCP moves to different positions with different orientations, at least three data are collected. Coupling with previously computed  $R_{sensor}$ , solving another set of linear equations leads to  $T_{sensor}$ .

Balls are preferred for the calibration, because the laser has a uniform view from any position and orientation [15]. In the proposed calibration algorithm, the position and the orientation of the 2-D planar disk do not affect the calibration. This enables the calibration target to be freely placed in the

system, such as the balls. During calibration,  $R_{sensor}$  and  $T_{sensor}$  are also solved separately with at least four conjugate pairs in total. Different from previous methods [11]–[15], our method collects more data  $\{T_{tcp,i}, R_{tcp,i}, q^i, D_i^i, D_i^b\}$  from one conjugate pair. First, data  $\{T_{tcp,i}, R_{tcp,i}, D_i^i, D_i^b\}$  are used to formulate (45) to derive  $R_{sensor}$ . Then, (47) is formulated using  $R_{sensor}$  and data  $\{T_{tcp,i}, R_{tcp,i}, q^i\}$  to obtain  $T_{sensor}$ . Sensing data are usually collected manually. Users are required to try different  $T_{tcp}$  and  $R_{tcp}$  values to locate different conjugate pairs. Smaller number of conjugate pairs makes the sensing data collection easier. The ball-based calibration methods [11]–[14] need to collect two sets of sensing data, which needs at least seven conjugate pairs in total. The first four conjugate pairs used for solving rotation portion required at least four  $T_{tcp}$  with the same  $R_{tcp}$ . In the second set of data for solving the translation portion, two  $T_{tcp}$  with the same  $R_{tcp}$  are required for each conjugate pair. Compared with the ball-based calibration, the proposed algorithm only requires one set of the sensing data with at least four conjugate pairs in total.

2) *Similar Calibration Accuracy*: The ball in Fig. 14(b) is used to illustrate that the proposed calibration method can achieve similar accuracy as the ball-based method does. In this comparison, the disk-based calibration and the ball-based calibration are based on four and eight conjugate pairs, respectively. Table III lists the eight sensing data  $\{T_{tcp,i}, R_{tcp,i}, q^i\}_{i=1}^8$  derived for the ball-based calibration, where the first four sensing data with the same orientation are used for solving  $R_{sensor}$  and the others with different orientations are used for deriving  $T_{sensor}$ . The ball-based calibration gives

$$\Upsilon = \begin{bmatrix} 0.99983 & -0.0115 & -0.01411 & 6.30066 \\ -0.01129 & -0.99983 & 0.01485 & 54.38647 \\ -0.01428 & -0.01468 & -0.99979 & 364.14357 \\ 0 & 0 & 0 & 1 \end{bmatrix}. \quad (53)$$

TABLE IV  
RECONSTRUCTED BALL CENTERS

|         | $\nu_i$ from Eq.(49)        | $\nu_i$ from Eq.(53)        |
|---------|-----------------------------|-----------------------------|
| $i = 1$ | (-321.193,-303.203,150.141) | (-321.068,-302.984,150.372) |
| $i = 2$ | (-320.712,-302.869,150.7)   | (-320.443,-302.813,150.796) |
| $i = 3$ | (-321.049,-302.838,150.485) | (-320.891,-302.673,150.707) |
| $i = 4$ | (-320.997,-302.84,150.181)  | (-320.878,-302.691,150.583) |
| $i = 5$ | (-320.746,-302.522,150.15)  | (-320.415,-302.855,150.755) |
| $i = 6$ | (-321.011,-302.605,150.254) | (-320.875,-302.618,150.433) |
| $i = 7$ | (-320.696,-302.671,150.477) | (-320.24,-302.909,150.836)  |
| $i = 8$ | (-320.945,-303.067,150.352) | (-320.754,-302.99,150.688)  |
| $i = 9$ | (-320.877,-302.896,150.311) | (-320.61,-302.785,150.665)  |

The deviation between (49) and (53) is

$$\Lambda^{-1}\Upsilon = \begin{bmatrix} 1.0000 & -0.0059 & -0.0070 & 0.2751 \\ 0.0059 & 0.9999 & -0.0119 & -0.2055 \\ 0.0071 & 0.0119 & 0.9999 & 0.1104 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (54)$$

with Euler angles

$$(0.0119, -0.0071, 0.0059). \quad (55)$$

Following Section III-D, the ball center from each orientation is:

$$\begin{bmatrix} \mu_i \\ 1 \end{bmatrix} = \begin{bmatrix} R_{\text{tcp},i} & T_{\text{tcp},i} \\ 0 & 1 \end{bmatrix} \Upsilon \begin{bmatrix} q^i \\ 1 \end{bmatrix}, \quad i \in \{5, 6, 7, 8\}.$$

The ball center is estimated as

$$\mu = \sum_{i=5}^8 \mu_i / 4$$

and the calibration accuracy is estimated as

$$\max_{i=5}^8 \|\mu - \mu_i\| = 0.208 \text{ mm}. \quad (56)$$

Equations (51) and (56) are similar. However, they are estimated according to different fixed points. In [23], calibration accuracy is evaluated as the standard deviation of the ball centers reconstructed from different scans. We follow the same way to compare the two calibration results (49) and (53). The scanning system scans the ball in Fig. 14(b) with the robotic arm in different orientations. In each scan, 50 profiles are sampled with the robotic arm in a fixed orientation. For  $i = 1, 2, \dots, 8$ , applying a transformation matrix [(49) or (53)] to the  $i$ th scan gives one point cloud  $C_i$ . On average, there are 45 000 points in each point cloud. The ninth point cloud is defined as  $C_9 = \bigcup_{i=1}^8 C_i$ . A ball is fitted to the point cloud  $C_i$  to derive the ball center  $\nu_i, i = 1, 2, \dots, 9$  (Table IV). The standard deviation of the ball centers is calculated as

$$\sum_{i=1}^8 \|\nu_i - \nu_9\| / 8.$$

The standard deviations for using (49) and (53) are 0.342 and 0.355 mm, respectively. Thus, we conclude that the disk-based calibration can provide a similar calibration result as the ball-based method does.

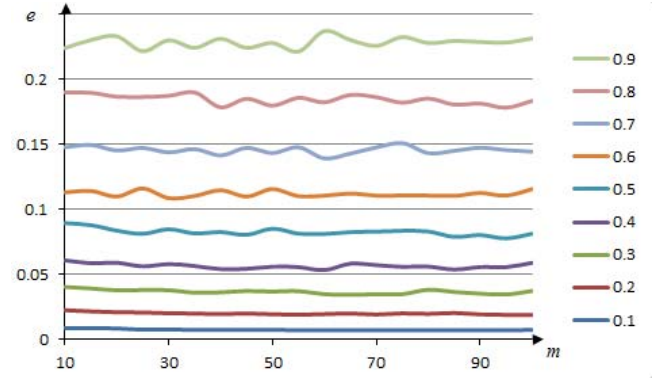


Fig. 17. Ball-based calibration accuracy with respect to  $\sigma$  and  $m$  [the horizontal coordinate is the number of arcs  $m$ ; the vertical coordinate is the calibration accuracy  $e(mm)$ , and curves in different colors correspond to different noise level  $\sigma(mm)$ ].

3) *Different Noise Tolerance*: The disk-based and ball-based calibrations are affected by different sources of noise. Disk-based calibration relies on the two tips on each chord, thus the  $x$ -resolution of the sensor affects the accuracy. On the other hand, the ball-based calibration depends on the arc data, which are affected by the  $z$ -accuracy of the sensor and the surface condition of the balls. The fitting algorithms adopted during the calibration may also introduce errors as well. Thus, instead of comparing their accuracies under the same noise level, we study their capability to tolerate a given noise. As presented in Fig. 15, increasing  $m$  is one way to tolerate the noise in disk-based calibration.

As presented in the last column of Table II, the ball-based calibration algorithms [11]–[14] collect one arc for each conjugate pair of  $R_{\text{sensor}}$ , and  $m$  arcs for each conjugate pair of  $T_{\text{sensor}}$ . Robotic data for deriving Table III are used to generate arcs. On each arc, points are generated in the sensor coordinate system with the  $x$ -resolution of 0.1 mm and Gaussian noise  $G(\sigma)$  on the  $z$  value of each point. For each  $m-\sigma$  pair, ball-based calibration is repeated for 500 times, and  $e$  is the standard deviation of the 500 ball centers reconstructed from the calibration. According to the simulation result in Fig. 17, in the same noise level  $\sigma$ , increasing  $m$  does not affect  $e$ , which is different from the disk-based calibration.

## VI. CONCLUSION

For calibrating profile sensors to robotic arms, all existing methods require 3-D calibration balls. Previously, 2-D disks are not used for robot-sensor calibration, because it is difficult to estimate its orientation during the calibration. In this paper, a new calibration algorithm is proposed to perform robot-sensor calibration based on a 2-D disk. The novel feature is that, in the calibration procedures, the 2-D disk can be placed in arbitrary orientation and this makes the placement of the disk as easy as that of a 3-D ball. To acquire the sensing data for our calibration algorithm, users can project the line laser onto the 2-D disk without passing any specific point. The proposed algorithm can derive the angular relationship between the 2-D disk and the laser's plane, and automatically obtain the disk center for the calibration. A new two-step calibration is proposed based on a single set of sensing



data. This can reduce not only the manual work in data collection but also the error propagation. Minimal manual work in data collection for the calibration is desired for better accuracy and faster processing. Simulations are performed to validate the proposed algorithm and analyze the effect of noise. Experiments show that the proposed algorithm can achieve similar accuracy as the ball-based method does.

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